Nagar Yuwak Shikshan Sansstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution Affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur.

Department Of Information Technology

ODD Term—2025-26 | Mid Semester Exam I

Semester – III | Date: -16/09/2025

Subject Code : **23IT1301** 1377493 | **Subject : Data structure and Program Design**

Time :1½ Hours | Max. Marks: 30

Instructions: Marks will be allotted as zero if wrong question number is mentioned.

Q.1. Solve any two	A) Write a program using structures and arrays to store information about employees (employee ID, name, and basic salary). Assume HRA = 20% of basic salary and DA = 10% of basic salary. Calculate the gross salary (Gross = Basic + HRA + DA) for each employee and display all details.	**[05M]**	CO1 (Level 1)
	B) Write a C program to read a string from the user and calculate the frequency of each character that appears in the string. Finally, display each character along with how many times it occurs in the string.	**[05M]**	CO1 (Level 1)
	C) Write a C program to input n numbers from the user. Use dynamic memory allocation (malloc) to store these numbers in an array. Then display the numbers in reverse order.	**[05M]**	CO1 (Level 2)
Q. 2 Solve any two	A) Develop a menu driven Program in C for the following operations on STACK. a. Push an Element on to Stack b. Pop an Element from Stack e. Display the status of Stack f. Exit	**[05M]**	CO1,CO2 (Level 2)
	B) Develop a menu driven Program in C for the following operations on QUEUE. a. Insert an Element on to QUEUE b. Delete an Element from QUEUE c. Demonstrate Overflow and Underflow situations on QUEUE d. Display the status of QUEUE e. Exit	**[05M]**	CO1,CO2 (Level 2)
	C) A hospital uses a circular queue to manage patient registration. Frame a menu-driven program that allows: • Registering a patient (Enqueue) • Serving a patient (Dequeue) • Displaying the waiting list.	**[05M]**	CO1,CO2 (Level 3)
Q.3 Solve any One	A) Write a menu-driven program to **create** a singly linked list. Add menu options to **insert a node** at: • The beginning • The end • A specific position	**[10M]**	CO1,CO2 (Level 2)
	B) Implement a menu-driven program-To **search** for an element in the linked list. To **count the total number of nodes** in the list. And To **display all elements** of the linked list.	**[10M]**	CO1,CO2 (Level 2)

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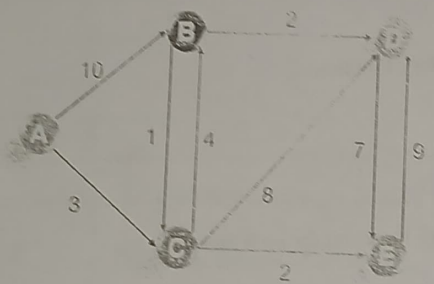
Hingna Road, Wanadongri, Nagpur.

Odd Term—2025-26 | Mid Semester Exam - II
Semester – III | Date: - 17/11/2025
Subject Code : 23IT1301 | Subject : Data Structure and Program Design
Time : 1½ Hours | Max. Marks: 30

Note:-
1) Attempt all questions.
2) Figures in bracket to the right indicate the marks for questions.
3) Marks will be allotted as zero if wrong question number is mentioned.

		Max Marks	CO Level
Que 1	**Solve any two of following**		
	(a) Insert the following keys into an initially empty Binary Search Tree in the order: 50, 25, 75, 22, 40, 60, 80, 90, 15, 30. After constructing the tree, draw the final BST, then write the Inorder, Preorder, and Postorder traversals of the tree, and finally write the Level Order traversal.	[5]	CO3 (Level 4)
	(b) Construct a B-Tree of order 5 by inserting the following keys in the given order: 1, 6, 7, 2, 11, 5, 10, 13, 12, 20, 16, 24, 3, 4, 18, 19, 14, 25. Start with an empty tree and show the B-Tree after each insertion, clearly indicating all node splits, and draw the final B-Tree.	[5]	CO3 (Level 4)
	(c) Insert the keys 6, 5, 4, 3, 2, 1 into an AVL tree. Show the AVL tree after each insertion, compute balance factors, and clearly identify all rotations performed to maintain balance. Draw the final AVL tree neatly.	[5]	CO3 (Level 4)

		Max Marks	CO Level
Que 2	**Solve any two of following**		
	(a) Obtain the following for the graph given below.	[5]	CO3 (Level 3)

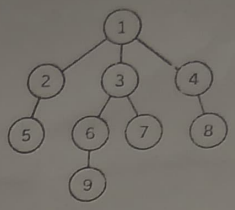


i) Indegree of each node
ii) Outdegree of each node
iii) Adjacency matrix
iv) Adjacency list

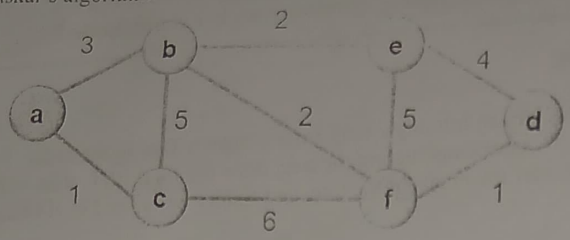
	(b) Perform a **Depth-First Search (DFS)** and a **Breadth-First Search (BFS)** on this graph shown below, both starting from vertex 1. For the DFS, show the order in which vertices are visited and the contents of the **stack** after each operation. For the BFS, show the visit order	[5]	CO3 (Level 3)

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and the contents of the **queue** after each step. Assume that adjacency is processed in increasing numerical order.



(c) What is Minimum Cost Spanning Tree? Obtain MST with its cost for given undirected graph using Kruskal's algorithm. [5] CO4 (Level 4)



Que 3 Solve any two of following

	Max Marks	CO Level
(a) Write a C program that copies the contents of one file into another. The program should prompt the user to enter the name of the source file and the name of the destination file, open the source file for reading and the destination file for writing, copy all data from the source to the destination, and display an appropriate message indicating whether the copy operation was successful or if either file could not be opened.	[5]	CO4 (Level 4)
(b) Insert the following keys into a hash table of size m = 10 using **open addressing:** **23, 43, 13, 27, 37, 18, 29**	[5]	CO4 (Level 4)

Use the primary hash function
$h(k) = k \bmod 10$
and, for double hashing, use the secondary function
$h_2(k) = 1 + (k \bmod 9)$.

Construct the hash table using:

1. **Linear Probing**
2. **Double Hashing**

Show the location of each key after every insertion for bo h methods.

(c) Write a C program that asks the user to enter the number of rows and columns of a matrix, reads the matrix elements, and then prints the transpose without creating a second matrix. The program should display the original matrix and then print the transpose by accessing elements in transposed order during output. [5] CO1 (Level 4)

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